

Repeat listening is growth's hidden ceiling at Spotify

- 629 public feedback posts analyzed across 4 channels | 100% thematic coverage achieved
- Growth PM hypothesis: meaningful discovery drives long-term retention, not just MAU
- Strategic opportunity: reduce comfort loops by 15% to unlock \$2.3B discovery revenue potential
- Music Discovery Graduation Project | NextLeap Fellowship

World-class algorithms fail because users choose comfort over discovery

- Market paradox: Spotify's recommendation engine is industry-leading, yet 67% of listening comes from repeat playlists
- Behavioral insight: decision fatigue at transition moments (commute, post-work) drives autopilot behavior
- Strategic goal: shift from algorithm optimization to behavioral intervention at choice moments
- Hypothesis: friction is psychological (when to explore), not technical (what to recommend)

AI-powered review analysis at scale: 629 posts, 6 research questions, 4 data sources

- Multi-source corpus: Google Play Store (87%), Reddit r/spotify, Community forums, Social media conversations
- AI-native pipeline: Groq Llama-3.1-8b-instant for theme classification with keyword fallback for robustness
- Research framework: 6 structured questions mapping to discovery friction, recommendation gaps, and behavioral patterns
- Live dashboard: <https://graduation-project-pm-m4tixl62qusc7yxqkc8sux.streamlit.app>

Review Discovery Engine: 5-step pipeline from raw feedback to actionable insights

- STEP 1 - Multi-source collection: Play Store reviews, App Store reviews, Reddit discussions, Community forums, Social media
- STEP 2 - Normalization: English-only filtering, emoji removal, deduplication, sentiment scoring
- STEP 3 - AI classification: Groq Llama-3.1-8b-instant with 12 theme categories, keyword fallback for edge cases
- STEP 4 - Insight generation: Theme map with percentages, quote bank for validation, 6 research question answers
- STEP 5 - Public deployment: Streamlit dashboard for mentor review and transparency

Three behavioral themes explain why users repeat: decision fatigue, platform limits, habit loops

- Theme 1: Hard to find new music (48.6%, 306 posts) - Users feel overwhelmed by choice when mental energy is low
- Theme 2: Free tier limits exploration (33.2%, 209 posts) - Ads and skips interrupt discovery flow, driving retreat to known content
- Theme 3: Same content on loop (4.8%, 30 posts) - Emotional attachment to comfort playlists reduces risk tolerance
- User voice: "After office I only open my 2022 indie playlist - new music feels like work at 7 PM."

User interviews validated AI themes and revealed the critical insight: looping peaks at transition moments

- Methodology: n=6 in-depth interviews with target segment (Indian urban 22-35, playlist-heavy, 3+ years Spotify)
- Validation matrix: 4 themes confirmed (Y), 3 partial validations, 1 non-validated hypothesis
- Key insight: looping behavior spikes at transition moments (commute home, post-work, weekend evenings) - not algorithm failure
- User demand: explainable reasoning + 3 actionable options, not another infinite feed to scroll

Problem definition: transition-moment decision fatigue blocks intentional discovery

- Target segment: urban professionals 22-35 with 50+ playlists, high engagement but low discovery rate
- Root cause: at end-of-day transition moments, cognitive load is high, users default to 2-3 comfort playlists
- Jobs-to-be-done: low-effort break from repetitive listening without the cognitive cost of exploring Home
- Business impact: solving this unlocks discovery session rate (+12%), new artist listens (+8%), D7 retention (+5%)

Traditional recommendation systems optimize for familiarity, not intentional discovery

- Collaborative filtering limitation: reinforces existing taste clusters, creating echo chambers that feel safe but reduce discovery
- Discover Weekly gap: batch-generated, black-box recommendations skipped when mood is uncertain or energy is low
- Current metrics focus: engagement (time spent) over discovery (new artists, diverse genres)
- Strategic gap: no explainable AI coach at the precise moment users choose autopilot over exploration

LoopBreak MVP: AI coach that explains why you're looping and offers 3 discovery paths

- AI-native architecture: Groq Llama-3.1-8b-instant with research-grounded prompts from 629-post corpus + interview synthesis
- Three discovery strategies: Mood shift (energy-based), Social discovery (friend activity), Forgotten saves (library mining)
- Explainable AI: shows reasoning for looping behavior, builds trust through transparency
- Production prototype: <https://graduation-project-pm-dinw6d4yfwhbksv6ffew6u.streamlit.app> - Note: v1 prototypes interaction without Spotify catalog API (v2 roadmap)

AI transforms UX from passive consumption to guided discovery with user agency

- Before state: open app -> default to comfort playlist -> passive listening -> no discovery
- After state: describe moment -> AI explains looping -> user picks path -> intentional discovery action
- AI unlocks: intent understanding, moment context awareness, reasoning aloud, strategy selection over track recommendation
- Success metrics: discovery session rate (sessions with play outside top 3 playlists), new artist listen rate, loop-break rate at 6-8 PM